

EXPLORING EMOTIONAL TERRAIN A Study of Sentiment Analysis in Textual Data

Amudalapalli Snehika¹ (snehika099@gmail.com)

Nimmakuri Chandu Tej¹ (chandutejnimmakuri@gmail.com)

Yalamarthy Samyuktha¹ (yalamarthisamyuktha2@gmail.com)

Tanikella Lakshmi Sowmya¹ (lakshmisowmya.tanikella@gmail.com)

Chinta Venkata Murali Krishna² (muralikrishnachinta2007@yahoo.co.in)

¹UG Students (CSE-DS), NRI Institute of Technology, Vijayawada, Andhra Pradesh, 521212

² Associate professor (CSE-DS), NRI Institute of Technology, Vijayawada, Andhra Pradesh, 521212

ABSTRACT:

The objective of the natural language processing (NLP) aspect of sentiment analysis is to automatically distil and aggregate sentiments from textual sources, such as online reviews and consumer feedback. This research proposes a methodology for categorizing Amazon food reviews on a scale of 1 to 5 based on expressed sentiments within the text, utilizing sentiment analysis techniques. Employing machine learning algorithms, the system discerns the overall sentiment conveyed in reviews by analyzing linguistic patterns within them. Data preprocessing techniques, including text cleansing, tokenization, and noise reduction, are applied to prepare the data for analysis. The textual data is then transformed into numerical representations using feature extraction methods such as bag-of-words and TF-IDF. Supervised learning algorithms, including logistic regression, support vector machines (SVM), and random forests, are trained on the pre-processed data to develop sentiment categorization models. The efficacy of these models is evaluated through pertinent metrics to determine the optimal approach. A pivotal aspect of this proposed methodology involves the presentation of sentiment analysis outcomes through word clouds, visually highlighting key terms and expressions influencing both positive and negative sentiment ratings. These visualizations offer businesses valuable insights into factors shaping consumer perceptions and preferences. Overall, this research equips businesses with a valuable tool to monitor and comprehend public sentiment regarding their food products, facilitating data-informed decision-making to enhance customer satisfaction and competitive positioning in the marketplace. The integration of NLP, machine learning, and visualization techniques demonstrates the system's capacity to extract actionable insights from extensive volumes of user-generated content.

Keywords: *Sentiment Analysis, Natural Language Processing (NLP), Amazon Food Reviews, Machine Learning, Text Preprocessing, Feature Extraction, Supervised Learning, Logistic Regression, Support Vector Machines (SVM), Customer Satisfaction.*

1. INTRODUCTION:

The web's explosive growth has produced a previously unheard-of volume of user-generated material, such as reviews, comments, and feedback from users. For companies looking to gauge public opinion about their goods and services, this data deluge offers both possibilities and difficulties. This problem is addressed by the natural language processing (NLP) area of sentiment analysis, which automates the extraction and aggregation of opinions from textual material on the internet. Sentiment research can offer priceless insights into consumer opinions on goods, flavours, packaging, and overall dining experiences in the food and grocery retail sector. An enormous amount of customer evaluations are produced every day due to the growth of e-commerce sites like Amazon, providing a wealth of data for sentiment analysis. Nevertheless, it is an intimidating and resource-intensive operation to manually sort through and comprehend these enormous amounts of textual data.

The goal of this project is to create an automated sentiment analysis system designed especially for Amazon food reviews. The suggested system would categorize customer reviews on a scale of 1 to 5, which represents sentiment polarity ranging from extremely negative to highly positive, by utilizing sophisticated natural language processing techniques and machine learning algorithms. Businesses will be able to determine areas for product improvement and measure client satisfaction levels thanks to this quantitative method to sentiment analysis. The word cloud-based sentiment analysis result display is a crucial part of the suggested system. These word clouds will draw attention to the word clusters that have the biggest impact on a review's overall mood rating. Businesses can obtain deeper insights into the elements influencing consumer perceptions and preferences—like taste, packaging, or ingredient quality—by visualizing the key words and phrases.

Comparing the suggested system to conventional sentiment analysis techniques reveals a number of benefits. Initially, it is made to manage the massive amount of data produced by Amazon's wide user base, allowing for quick decision-making and real-time analysis. Second, by employing cutting-edge NLP methods and machine learning models, sentiment classification is highly accurate, lowering the possibility of misinterpreting client comments. Third, the word cloud visualizations help data-driven decision-making processes by offering a clear and understandable depiction of the underlying sentiment drivers. Businesses in the food sector can obtain a competitive edge by better understanding and catering to client preferences by utilizing the enormous volume of user-generated material on Amazon. Within the food and grocery retail industry, the suggested sentiment analysis system has the power to completely change how companies handle customer engagement, marketing tactics, and new product creation.

2. LITERATURE REVIEW

The sentiment analysis of online reviews, particularly in the domain of e-commerce platforms like Amazon, has been a subject of growing interest among researchers. Hague, Saber, and Shah (2018) conducted a study focusing on sentiment analysis of large-scale Amazon product reviews. Their research provided insights into the application of sentiment analysis techniques on vast datasets, highlighting the challenges and opportunities in extracting meaningful sentiments from a plethora of user-generated content.

Almjawel et al. (2019) delved into sentiment analysis and visualization of Amazon books' reviews, shedding light on the importance of visualizing sentiment trends and patterns to facilitate data interpretation and decision-making processes. Their study emphasized the role of visualization techniques in enhancing the understanding of consumer sentiments and preferences regarding specific product categories.

In a study by Shyamasundar and Rani (2016), Twitter sentiment analysis was explored using different feature extractors and dimensionality reduction techniques with supervised learning algorithms. Their research contributed to the understanding of various feature extraction methods and their impact on sentiment classification accuracy, providing valuable insights for researchers and practitioners in the field of sentiment analysis.

Shalini et al. (2018) focused on sentiment analysis for code-mixed Indian social media text, employing distributed representation techniques. Their research addressed the challenges posed by multilingual and code-mixed text data, offering novel approaches for sentiment analysis in linguistically diverse contexts.

Yadlapalli, Reddy, and Sasikala (2020) conducted advanced Twitter sentiment analysis using supervised techniques and minimalistic features. Their study demonstrated the effectiveness of supervised learning algorithms in capturing nuanced sentiments expressed in tweets, highlighting the importance of feature selection and model optimization in sentiment analysis tasks.

Overall, the literature reviewed underscores the diverse methodologies and techniques employed in sentiment analysis tasks, ranging from large-scale e-commerce platforms like Amazon to social media platforms such as Twitter. These studies contribute to the advancement of sentiment analysis research, offering valuable insights and methodologies for analyzing user-generated content and extracting meaningful sentiments.

3. PROPOSED WORK:

This research proposes a comprehensive approach to sentiment analysis of food reviews. With the rise of e-commerce platforms and the abundance of user-generated content, analyzing sentiment from reviews has become increasingly important for businesses to understand customer opinions and improve their products or services. A number of crucial processes are involved in the suggested model for sentiment analysis of Amazon food reviews: data collection, data preprocessing, data splitting, feature extraction and selection, and model evaluation.



Fig 1: Architecture of Proposed System

3.1. Data Collection:

This case study focusses on the *fine food review data set on amazon* which is available on Kaggle. The data set consists of reviews of fine foods from amazon over a period of more than 10 years, including 568,454 reviews till October 2012. Reviews include rating, product and user information, and a plain text review. It also includes reviews from all other Amazon categories.

3.2. Data Preprocessing:

The data goes through a series of meticulous data preprocessing techniques are employed to ensure the text data is optimally prepared for sentiment analysis. Firstly, missing values are addressed through the removal of any rows containing null entries. This ensures the dataset remains pristine and devoid of any incomplete records that could compromise analysis outcomes. Following this, duplicate entries are identified and eliminated, safeguarding against skewed results and biased model training.

Secondly, a comprehensive text cleaning process is enacted to refine the textual content. This involves the removal of extraneous elements such as punctuation, special characters, and stop words, ensuring that the text is devoid of noise and irrelevant information. Moreover, feature engineering techniques are applied to create new informative features, augmenting the dataset with additional insights. Conversions of Unix time to human-readable date formats further enhance the dataset's interpretability, facilitating temporal analysis. Overall, these meticulous preprocessing steps guarantee that the text data is refined, standardized, and enriched.

3.3. Exploratory Data Analysis (Eda):

Exploratory Data Analysis (EDA) guides the explicit addition of new features aimed at enriching the dataset's informational depth. Notably, the creation of the *'Review_length'* feature through histogram visualization provides insights

into the distribution of review lengths, offering a nuanced understanding of the textual data's length variation. This addition facilitates the identification of potential trends or outliers in review lengths, which could inform subsequent analysis or modelling decisions.

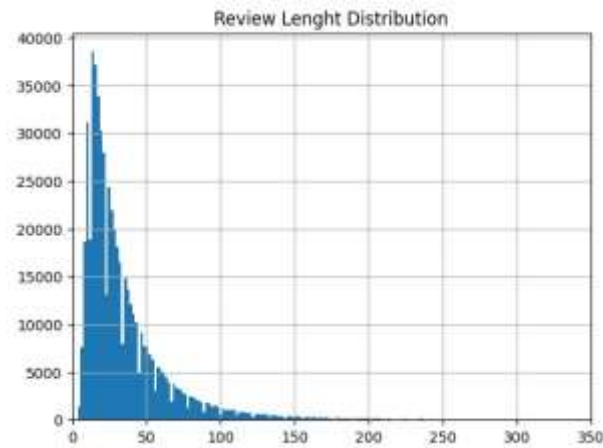


Fig: Distribution of length of reviews

Additionally, the 'Score' feature is strategically visualized through a countplot, elucidating the distribution of review scores across different sentiment classes. This visualization offers valuable insights into the distribution of sentiment labels within the dataset, laying the foundation for sentiment analysis modelling by highlighting the prevalence of positive, negative, and neutral sentiments.

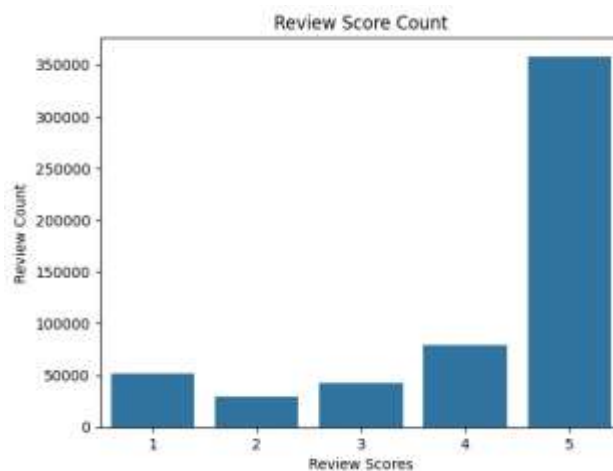


Fig: Bar chart for count of each rating classification

3.4. Sentiment Analysis:

Sentiment analysis is conducted to categorize food reviews into three sentiment classes: positive, neutral, and negative. This sentiment analysis is based on the review scores associated with each review. A custom function `score_classification` is defined to classify the review scores into these sentiment categories: scores greater than 3 are labelled as positive(1), scores equal to 3 are labelled as neutral(2), and scores less than 3 are labelled as negative(0). These sentiment labels are then assigned to a new column called 'Sentiment' in the dataset.

Subsequently, a visualization, such as a countplot, is utilized to depict the distribution of sentiment labels across the dataset, providing insights into the prevalence of positive, neutral, and negative sentiments among the food reviews. Through this sentiment analysis process, the code aims to categorize and understand the sentiment polarity of the food reviews, laying the groundwork for further analysis and modelling tasks.

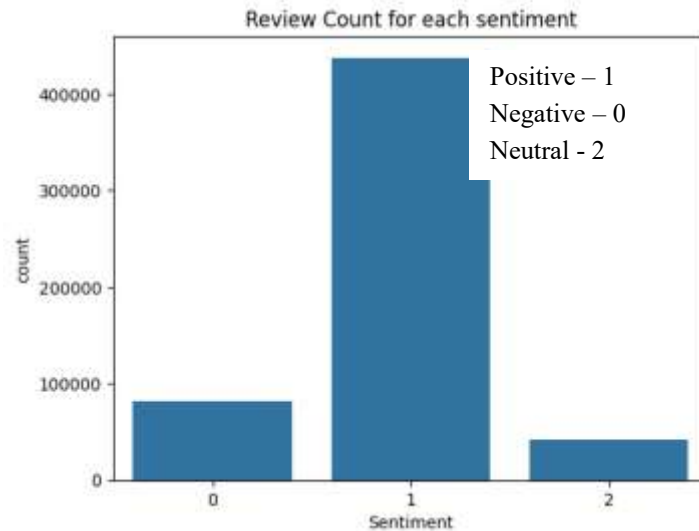


Fig: Visualization of Positive negative and neutral review count

3.5. Model Building:

3.5 TF-IDF:

TF-IDF vectorization plays a crucial role in the project by serving as the foundation for feature engineering. The resulting TF-IDF matrices (X_{train_tfidf} and X_{test_tfidf}) act as the input features for various sentiment analysis models such as Logistic Regression, Support Vector Machine (SVM), and Random Forest. By transforming raw text into numerical representations, TF-IDF enables the models to capture semantic information and discern patterns related to sentiment polarity more effectively.

3.5.1 Logistic Regression:

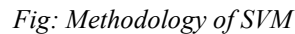
Logistic Regression serves as one of the primary classification algorithms utilized for sentiment analysis in the code. Logistic Regression models the probability of a binary outcome (positive or negative sentiment) based on input features, such as TF-IDF vectors representing the textual content of reviews.

$$P(y=1|X) = 1 / (1 + e^{(-z)})$$

The logistic function, also known as the sigmoid function, is applied to the linear combination of input features and model parameters to produce probabilities. Through iterative optimization techniques like gradient descent, the model learns the optimal coefficients to maximize the likelihood of observing the actual sentiment labels in the training data. This approach enables Logistic Regression to effectively classify reviews into positive, neutral, or negative sentiment categories based on their TF-IDF representations.

3.5.2 SVM:

Support Vector Machine (SVM) is another powerful classification algorithm leveraged for sentiment analysis in the code. SVM aims to find the optimal hyperplane that best separates the data points of different sentiment classes in the feature space defined by TF-IDF vectors.



3.5.3 Random Forest:

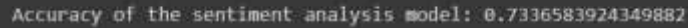
3.6. Evaluation Models:

4. RESULTS:

[illegible][illegible]

Page No: 90

4.2 Logistic Regression model result:



Accuracy of the sentiment analysis model: 0.7336583924349882

Fig: Accuracy of Logistic Regression Model

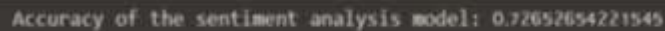
4.3 SVM model result:



Accuracy of the sentiment analysis model: 0.68839457896352

Fig: Accuracy of SVM Model

4.4 Random Forest result:



Accuracy of the sentiment analysis model: 0.72652654221545

Fig: Accuracy of Random Forest Model

5. FUTURE SCOPE:

Looking ahead, there is considerable scope for advancing and refining the sentiment analysis project. Future enhancements could include exploring advanced text processing techniques such as word embeddings and deep learning architectures for improved sentiment analysis performance. Additionally, fine-grained sentiment analysis, aspect-based sentiment analysis, and integration of multimodal data sources could offer more nuanced insights and cater to diverse application scenarios.

Furthermore, the project could be extended to real-time sentiment analysis and adaptation to different domains, enabling businesses to monitor sentiment trends and make informed decisions in a timely manner. Continuous evaluation and refinement of sentiment analysis models would be essential to ensure robust performance and applicability across various contexts and industries.

6. CONCLUSION:

The sentiment analysis research on Amazon food reviews offers valuable insights into customer attitudes and opinions regarding the range of food products available on the platform. By employing machine learning algorithms and natural language processing techniques, the study effectively categorizes reviews into positive, negative, and neutral sentiments, based on the expressed thoughts of consumers. Analyzing a dataset of approximately five lakh Amazon food review records, the research underscores the significance of sentiment analysis in understanding consumer behaviour and preferences. Through data preprocessing, including duplicate handling, feature engineering, and text cleaning, the dataset was prepared for analysis, ensuring the accuracy and relevance of sentiment analysis findings. Utilizing random forest, logistic regression, and support vector machines (SVM), the sentiment analysis model accurately predicts sentiment labels for reviews, providing stakeholders and businesses with actionable insights. Additionally, word clouds and frequency distributions highlight the most commonly used terms associated with positive and negative attitudes, offering a comprehensive understanding of factors influencing consumer perceptions. Overall, this study emphasizes the importance of sentiment analysis in enhancing decision-making processes within the food industry and underscores its value in extracting insightful information from vast volumes of customer-generated data.

7. REFERENCES:

- 1 T. U. Hague, N. N. Saber and F. M. Shah, "Sentiment analysis on large scale Amazon product reviews", 2018 IEEE International Conference on Innovative Research and Development (ICIRD), 2018.
- 2 Almjawel, S. Bayoumi, D. Alshehri, S. Alzahrani and M. Alotaibi, "Sentiment Analysis and Visualization of Amazon Books' Reviews", 2019 2nd International Conference on Computer Applications & Information Security (ICCAIS), 2019.
- 3 CVM Krishna, GA Rao, S AnuRadha, KVD Sagar, "Prediction of User Overall Gratification in Indian Tourism Domain on Hotel Classes and Trip Types", ECS Transactions 107 (1), 19813
- 4 L. B. Shyamasundar and P. J. Rani, "Twitter sentiment analysis with different feature extractors and dimensionality reduction using supervised learning algorithms", 2016 IEEE Annual India Conference (INDICON), 2016.
- 5 K. Shalini, H. B. Ganesh, M. A. Kumar and K. P. Soman, "Sentiment Analysis for Code-Mixed Indian Social Media Text With Distributed Representation", 2018 International Conference on Advances in Computing, Communications and Informatics (ICACCI), pp. 1126-1131, 2018.
- 6 CVM Krishna, KB Sri, S Avinash, DA Priya, G Yamini, "Users preference assessment on overall gratification in asiantourism hotels before and after pandemic", AIP Conference Proceedings 2796 (1)
- 7 Saisrihitha Yadlapalli, R. Reddy and T. Sasikala, Advanced Twitter Sentiment Analysis Using Supervised Techniques and Minimalistic Features, 2020.
- 8 CVM Krishna, AC Sri, S Ayesha, GR Lakshmi, DP Shankar, "Impact of user reviews and ratings on overall rating in Indian tourism hotels", AIP Conference Proceedings 2796 (1).
- 9 T. U. Hague, N. N. Saber and F. M. Shah, "Sentiment analysis on large scale Amazon product reviews", 2018 IEEE International Conference on Innovative Research and Development (ICIRD), 2018.
- 10 A. Almjawel, S. Bayoumi, D. Alshehri, S. Alzahrani and M. Alotaibi, "Sentiment Analysis and Visualization of Amazon Books' Reviews", 2019 2nd International Conference on Computer Applications & Information Security (ICCAIS), 2019.
- 11 L. B. Shyamasundar and P. J. Rani, "Twitter sentiment analysis with different feature extractors and dimensionality reduction using supervised learning algorithms", 2016 IEEE Annual India Conference (INDICON), 2016.
- 12 K. Shalini, H. B. Ganesh, M. A. Kumar and K. P. Soman, "Sentiment Analysis for Code-Mixed Indian Social Media Text With Distributed Representation", 2018 International Conference on Advances in Computing, Communications and Informatics (ICACCI), pp. 1126-1131, 2018.
- 13 CVM Krishna, GA Rao, S Anuradha, "Analysing the impact of contextual segments on the overall rating in multi-criteria recommender systems", Journal of big Data 10 (1), 16
- 14 Saisrihitha Yadlapalli, R. Reddy and T. Sasikala, Advanced Twitter Sentiment Analysis Using Supervised Techniques and Minimalistic Features, 2020.